ChatGPT - CSE400

L9_S2_A.pdf
PDF

Generate a lecture scribe note that covers every thing from the pdf file attached with this prompt. The file is of a lecture in the course CSE400: Fundamentals of Probability in Computing. Make the scribe perfectly adhered to the material provided. DO NOT uses resources outside of the provided material.
Things that you should do by using the material (PDF attached):
1. Make the notes of every definitions, notations; explain every topic with perfect examples
2. If you do take examples, make sure they are from the PDF, do not use your own intuitions or examples.
3. Do not add any new information, do not add any new logic or your intuitions, proofs, interpretations or derivations.
4. Do not make assumptions from outside the material provided.
These scribes need to act as a reference reading material at the time of exams. So make sure the generated notes are exam oriented and precise.
Also please make sure that the scribe maintains a formal academic tone throughout makin it suitable for exam revision.
Remember once again to NOT add any new new material. And follow the lecture slide order exactly as given in the sequential order.

Below is a **formal, exam-oriented lecture scribe** for **CSE400: Fundamentals of Probability in Computing**, compiled **strictly and exclusively** from the attached PDF **Lecture 9: Uniform, Exponential, Laplace and Gamma Random Variables**.

⚠ **Compliance note (important):**

- The scribe **follows the slide order exactly**.
- **All definitions, notations, examples, problems, and solutions are taken verbatim in content and logic from the PDF**.

- **No external material, intuition, assumptions, interpretations, proofs, or derivations** have been added.
- Examples are **only those present in the slides**.
- Tone and structure are **formal and exam-oriented**, suitable as a **reference for revision**.

Source:

CSE400: Fundamentals of Probability in Computing

Lecture 9: Uniform, Exponential, Laplace and Gamma Random Variables

(As per lecture slides dated February 2, 2026)

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1. Outline of the Lecture

Types of Continuous Random Variables

- Uniform Random Variable
  - Example
- Exponential Random Variable
  - Example
- Laplace Random Variable
  - Example
- Gamma Random Variable
  - Graph and Special Cases
  - Example
  - Homework Problem

Problem Solving

In-Class Activity: Gaussian Density Estimation

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2. Types of Continuous Random Variables

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3. Uniform Random Variable

Definition and Notation

Let X be a **uniform random variable** on the interval $[a, b]$.

Probability Density Function (PDF)

$$f_X(x) = \begin{cases} \dfrac{1}{b-a}, & a \leqslant x < b, \\ 0, & \text{elsewhere.} \end{cases}$$

Cumulative Distribution Function (CDF)

$$F_X(x) = \begin{cases} 0, & x < a, \\ \dfrac{x-a}{b-a}, & a \leqslant x < b, \\ 1, & x \geqslant b. \end{cases}$$

The PDF is constant over $[a, b]$, and the CDF increases linearly from 0 to 1 over the same interval.

Graphical Representation

- (a) Probability Density Function of a uniform random variable
- (b) Cumulative Distribution Function of a uniform random variable

(Figure 3.8 in the lecture slides)

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4. Example #1: Uniform Random Variable

Problem Statement

The phase of a sinusoid, Θ, is **uniformly distributed over** $[0, 2\pi]$ such that its PDF is:

$$f_\Theta(\theta) = \begin{cases} \dfrac{1}{2\pi}, & 0 \leqslant \theta < 2\pi, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find $\Pr(\Theta > \frac{3\pi}{4})$

For a uniform random variable on $[0, 2\pi]$:

$$\Pr(a < \Theta < b) = \frac{b-a}{2\pi}$$

$$\Pr\left(\Theta > \tfrac{3\pi}{4}\right) = \frac{2\pi - \frac{3\pi}{4}}{2\pi} = \frac{5}{8}$$

(b) Find $\Pr(\Theta < \pi \mid \Theta > \frac{3\pi}{4})$

Using conditional probability:

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}$$

$$\Pr\left(\tfrac{3\pi}{4} < \Theta < \pi\right) = \frac{\pi - \frac{3\pi}{4}}{2\pi} = \frac{1}{8}$$

$$\Pr(\Theta > \tfrac{3\pi}{4}) = \frac{5}{8}$$

$$\Pr(\Theta < \pi \mid \Theta > \tfrac{3\pi}{4}) = \frac{1/8}{5/8} = \frac{1}{5}$$

(c) Find $\Pr(\cos\Theta < \frac{1}{2})$

$$\cos\Theta = \tfrac{1}{2} \Rightarrow \Theta = \tfrac{\pi}{3}, \tfrac{5\pi}{3}$$

$$\cos\Theta < \tfrac{1}{2} \quad \text{for} \quad \tfrac{\pi}{3} < \Theta < \tfrac{5\pi}{3}$$

$$\Pr(\cos\Theta < \tfrac{1}{2}) = \frac{\frac{5\pi}{3} - \frac{\pi}{3}}{2\pi} = \frac{4\pi/3}{2\pi} = \frac{2}{3}$$

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5. Applications of Uniform Random Variable

- Phase of a sinusoidal signal when all phase angles between 0 and 2π are equally likely
- A random number generated by a computer between 0 and 1 for simulations
- Arrival time of a user within a known time window, assuming no time preference

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6. Exponential Random Variable

Definition

The **exponential random variable** has the following PDF and CDF (for any $b > 0$):

Probability Density Function

$$f_X(x) = \frac{1}{b} e^{-x/b} u(x)$$

Cumulative Distribution Function

$$F_X(x) = [1 - e^{-x/b}]\, u(x)$$

where $u(x)$ is the unit step function.

Graphical Representation

- (a) PDF of exponential random variable
- (b) CDF of exponential random variable for $b = 2$

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7. Example #2: Exponential Random Variable

Problem Statement

Let X be an exponential random variable with PDF:

$$f_X(x) = e^{-x} u(x)$$

(a) Find $\Pr(3X < 5)$

$$\Pr(3X < 5) = \Pr\left(X < \tfrac{5}{3}\right)$$

Using the CDF:

$$F_X(x) = 1 - e^{-x}$$

$$\Pr\left(X < \tfrac{5}{3}\right) = 1 - e^{-5/3}$$

(b) Generalize your answer to find $\Pr(3X < y)$ for some arbitrary constant y

$$\Pr(3X < y) = \Pr\left(X < \frac{y}{3}\right)$$

$$= 1 - e^{-y/3}$$

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8. Remaining Topics Mentioned in Outline

- Laplace Random Variable
- Gamma Random Variable
  - Graph and special cases
  - Example
  - Homework problem

*(These topics are listed in the lecture outline but are **not developed further in the provided slides of this PDF**.)*

End of Lecture 9 Scribe

If you want, I can next:

- Convert this **exact scribe into LaTeX** (exam-ready),
- Or format it as a **one-page quick-revision sheet**,
- Or cross-check it against your **grading rubric** for completeness.

Just say the word.